ψ-Field Quantum Control and Error-Suppression Architectures

Inventor: Gary Thomas Alcock

October 11, 2025

Cross-Reference to Related Applications

Pursuant to 35 U.S.C. §119(e), this application claims the benefit of the following U.S. provisional applications, the entirety of each being incorporated herein by reference:

- U.S. Provisional App. No. **63/865,279** (Confirmation No. 5785), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Gravitational and Signal Correction.*
- U.S. Provisional App. No. **63/865,283** (Confirmation No. 5905), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Energy Transmission.*
- U.S. Provisional App. No. **63/865,293** (Confirmation No. 6649), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Information Processing, Communication, and Advanced Computing.*
- U.S. Provisional App. No. **63/868,073** (Confirmation No. 8739), filed **Aug. 21, 2025**, entitled *Methods, Systems, and Apparatus for Artificial Generation, Control, and Application of Scalar Density Fields.*
- U.S. Provisional App. No. **63/893,633** (Confirmation No. 5179), filed **Oct. 4, 2025**, entitled *Systems and Methods for Electromagnetic Coupling, Actuation, and Application of Scalar Refractive Fields.*

This application also incorporates by reference, in their entirety for all purposes, **all other provisional and non-provisional applications filed by the same inventor on or after August 15, 2025**, including any continuations, divisionals, or continuations-in-part thereof. The foregoing applications are incorporated by reference in their entirety for all purposes, including definitions, figures, and embodiments. To the extent of any inconsistency, the present disclosure controls.

Statement Regarding Federally Sponsored Research or Development

No federal funding was used in connection with the inventions described herein. No government rights apply.

Incorporation by Reference (Additional Materials)

In addition to the applications identified above, technical materials authored by the inventor concerning scalar refractive field (ψ) dynamics, cavity–atom LPI tests, navigation/timing corrections, energy coupling, communication architectures, and quantum control are incorporated by reference. All essential material for the claimed inventions is included explicitly herein; incorporated materials are provided for background and alternative implementations.

Definitions; Conventions

As used herein, "comprising" is open-ended and does not exclude additional elements or steps; "including" and "having" are used in a like manner. Numerical ranges include all sub-ranges and individual values unless otherwise specified. Singular terms include the plural and vice versa unless context dictates otherwise. Functional descriptions are supported by the corresponding structures, algorithms, and examples disclosed herein; no limitation to "means-plus-function" under 35 U.S.C. §112(f) is intended unless expressly recited. Throughout, the scalar refractive field is denoted ψ, with optical index $n = e^{\psi}$ where applicable.

No Admission

Any discussion of references, techniques, or prior work in the Background is provided solely for context and is not an admission that any such material constitutes prior art, is known, conventional, or required.

Abstract

Disclosed are systems and methods for coherent control of quantum and sub-quantum systems using engineered scalar refractive fields (ψ) to stabilize phase, suppress decoherence, and perform error-corrective operations in hybrid EM–ψ architectures. By embedding qubits, resonators, or photonic elements in a controlled ψ potential, phase noise and energy dispersion are actively reduced through refractive-index modulation synchronized to system dynamics. The architecture unifies ψ-based feedback, parametric coupling, and synthetic gauge control to achieve fault-tolerant operation of qubits, cavities, or interferometers. Applications include quantum processors, clocks, sensors, and metrology platforms leveraging ψ stabilization.

Field of the Invention

The invention relates to quantum control, decoherence suppression, and precision measurement. More particularly, it concerns generation and control of a scalar refractive field (ψ) for stabilizing phase-coherent quantum systems and implementing active error-suppression protocols.

Background

Quantum devices are limited by decoherence, noise, and drift in environmental parameters such as electromagnetic fields, temperature, and mechanical vibrations. Existing mitigation relies on cryogenics, pulse shaping, or software-level error correction, each incurring cost or latency. There is no known architecture providing *field-level coherence stabilization* that operates continuously and locally. The ψ framework enables modulation of the effective refractive potential

$n = e^{\psi}$ experienced by quantum wavefunctions, thereby controlling phase and energy dispersion directly at the physical substrate.

Summary

A ψ-quantum controller generates and modulates a scalar refractive field overlapping one or more quantum devices. Dynamic ψ envelopes synchronize with qubit evolution or cavity cycles to null phase drift and energy spread. A feedback system measures quantum-state error syndromes and adjusts the ψ waveform accordingly, forming a continuous, analog layer of error suppression below conventional digital correction.

Representative Claims

1. Apparatus for quantum coherence stabilization, comprising: (i) a quantum device having at least one state sensitive to phase or energy noise; (ii) a ψ generator configured to apply a controllable scalar refractive field overlapping the quantum device; and (iii) a controller configured to modulate amplitude or phase of the ψ in synchrony with state evolution of the quantum device so as to reduce decoherence or phase drift.
2. Method for field-level error suppression, comprising: (a) monitoring one or more quantum error indicators (e.g., Ramsey fringes, Bloch-vector tilt); (b) computing a corrective ψ envelope minimizing variance of a target observable; and (c) applying the corrective field in real time to restore coherence.
3. Hybrid EM–ψ architecture, comprising coupled electromagnetic and scalar resonators driven with phase-locked modulation, whereby cross-coupling produces an effective synthetic gauge field that stabilizes multi-qubit entanglement.

Brief Description of the Drawings

Fig. 1: Concept of ψ-stabilized qubit system.
Fig. 2: Hybrid EM–ψ control loop architecture.
Fig. 3: Bloch-sphere error suppression illustration.
Fig. 4: Integrated multi-qubit module layout.

Detailed Description

Definitions

The scalar refractive field ψ determines an effective optical index $n = e^{\psi}$ and modifies the local phase velocity $c_1 = c\,e^{-\psi}$. When coupled to a quantum system with wavefunction $\Psi(\mathbf{r}, t)$, gradients $\nabla\psi$ act as refractive potentials that shift or stabilize phase evolution.

Qubit Stabilization Principle

A qubit evolving under Hamiltonian H_0 experiences environmental noise $\delta H(t)$. Applying a ψ modulation $\psi(t) = \psi_0 + \psi_1 \cos(2\pi f_m t + \phi)$ adjusts the effective index to counteract $\delta H(t)$, maintaining constant energy spacing $\hbar\omega_q$. Feedback sensors detect phase error and retune ψ_1 and ϕ.

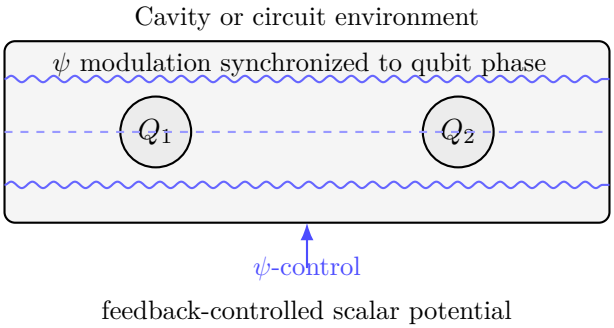

Figure 1: ψ-stabilized qubits within a cavity: the scalar field envelope modulates refractive index to counteract phase noise and maintain coherence.

Hybrid EM–ψ Control Architecture

The control system integrates EM sensing and ψ generation. Phase detectors extract error signals, which drive an adaptive controller computing corrective ψ envelopes.

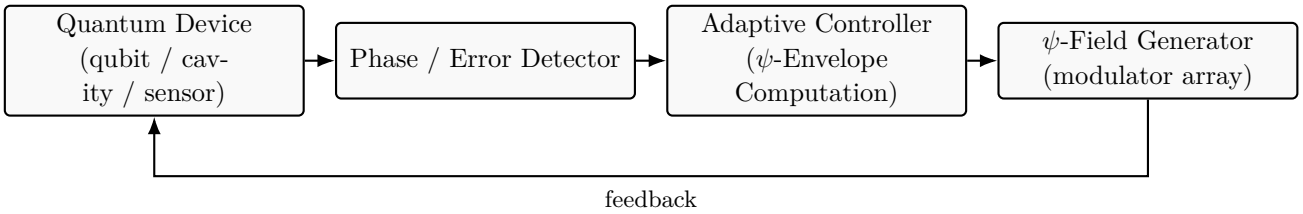

Figure 2: Hybrid EM–ψ control loop: measured phase error drives adaptive computation of corrective ψ waveforms for real-time stabilization.

Error-Suppression Dynamics

The ψ control acts continuously, replacing discrete digital correction with analog phase cancellation. Figure 3 illustrates reduced dephasing on the Bloch sphere.

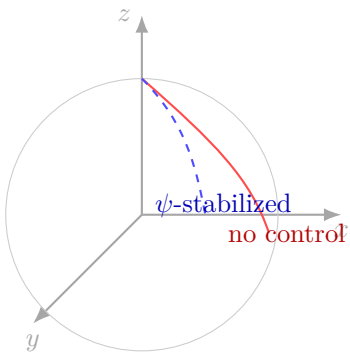

Figure 3: Bloch-sphere comparison: uncontrolled dephasing (red) vs. ψ-stabilized evolution (blue), showing suppression of phase diffusion.

Integrated Multi-Qubit Module

Multiple qubits can share a single ψ envelope distributed through a metamaterial substrate for correlated stabilization.

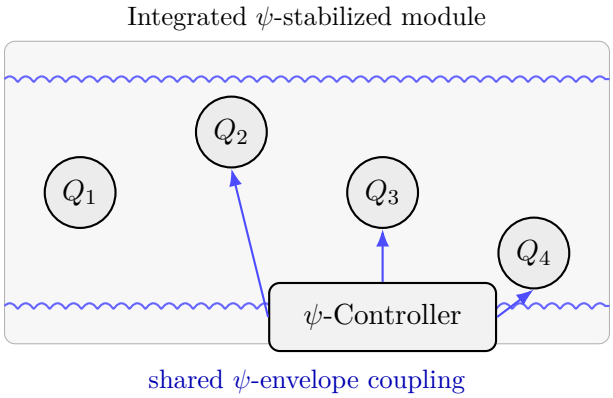

Figure 4: Integrated ψ-field control for multiple qubits: a shared scalar envelope stabilizes all qubits on a common substrate, supporting entanglement fidelity.

Alternative Embodiments

Embodiments include: (i) solid-state qubits in superconducting or spin platforms; (ii) optical interferometers with ψ-field phase compensation; (iii) atomic or ionic traps with refractive stabilization; (iv) continuous-wave ψ-feedback integrated with digital quantum-error-correction layers.

Industrial Applicability

Applicable to quantum computing, quantum networking, time-frequency metrology, and precision navigation systems requiring ultra-stable phase coherence.

Conclusion

By embedding quantum hardware in an actively controlled scalar refractive field, the disclosed architectures provide continuous, physical-layer suppression of decoherence and phase noise, forming a foundation for next-generation quantum devices and ψ-field–stabilized metrology.